Arista Access Node HEB-ACC-11 QoS Congestion Event Troubleshooting SOP

Node: HEB-ACC-11 | **Vendor:** Arista | **Severity:** Critical | **Alarm Code:** ALARM_4887

Description

On the Arista Access Node HEB-ACC-11, QoS congestion events (Microbursts, interface tail drops, QoS queue starvation) can arise due to the misconfigured Quality of Service (QoS) policies in the egress direction, which may not be properly accounting for the bandwidth requirements of the various classes of service. This can lead to a situation where high-priority traffic is being unnecessarily queued behind low-priority traffic, causing congestion on the interface. Furthermore, the lack of adequate policing and shaping mechanisms may allow bursts of high-bandwidth traffic to overwhelm the interface, causing microbursts and interface tail drops. Additionally, the QoS queue starvation issue can arise due to the insufficient buffer size allocated to the QoS queues, leading to packet drops and subsequent congestion.

Pre-requisite Checks

- Power cycle the affected node to ensure that there are no transient issues.

- Verify that all redundant links are operational and that there are no issues with the redundant path.

- Check the site's environmental conditions to ensure that they are within the recommended operating parameters.

Potential Root Causes

Cause	Probability
Misconfigured QoS policies in the egress direction	High
Lack of policing and shaping mechanisms	Medium
Insufficient buffer size allocated to QoS queues	Low

Diagnostic Commands & Expected Outputs

```
Command: show hardware qos queue statistics, Expected Output: Queue statistics for the egress direction, including the number of packets dropped due to congestion., Description: This command will provide insight into the congestion levels on the interface, helping to identify the root cause of the issue.

Command: show policer statistics, Expected Output: Policer statistics, including the number of packets and bytes dropped due to policing., Description: This command will help identify if there are any policing issues that may be contributing to the congestion.

Command: show qos queue buffer size, Expected Output: Buffer size allocated to each QoS queue, including the number of packets dropped
```

```
due to queue starvation., Description: This command will provide
insight into the buffer size allocated to the QoS queues, helping to
identify if queue starvation is a contributing factor.
```

Resolution Steps

Step 1: Step: Verify and adjust QoS policies, Command: configure terminal policy-map type service qos-policy class class-default police 1000000000 apply policy-map qos-policy to interface 1/1/1, Description: Verify that the QoS policies in the egress direction are correctly configured and adjust as necessary to ensure that high-priority traffic is not being unnecessarily queued behind low-priority traffic.

Step 2: Step: Implement policing and shaping mechanisms, Command: configure terminal policer policer-name cir 1000000000 apply policer policer-name to interface 1/1/1, Description: Implement policing and shaping mechanisms to prevent bursts of high-bandwidth traffic from overwhelming the interface and causing congestion.

Step 3: Step: Increase buffer size allocated to QoS queues, Command: configure terminal interface 1/1/1 queue buffer-size 1000000 apply running-config, Description: Increase the buffer size allocated to the QoS queues to prevent queue starvation and packet drops.

Post-Check Verification

```
Command: show hardware qos queue statistics, Expected Output: Queue
statistics for the egress direction, including the number of packets
dropped due to congestion., Description: Verify that the congestion
levels on the interface have decreased after implementing the
resolution steps.

Command: show policer statistics, Expected Output: Policer statistics,
including the number of packets and bytes dropped due to policing.,
Description: Verify that the policer statistics have decreased after
implementing the resolution steps.

Command: show qos queue buffer size, Expected Output: Buffer size
allocated to each QoS queue, including the number of packets dropped
due to queue starvation., Description: Verify that the buffer size
allocated to the QoS queues has increased after implementing the
resolution steps.
```

Roll-back Procedures

- Step: Undo policing and shaping mechanisms, Command: configure terminal delete policer policer-name apply running-config, Description: Undo the policing and shaping mechanisms to prevent any potential impact on the network.

- Step: Restore original buffer size allocated to QoS queues, Command: configure terminal interface 1/1/1 undo queue buffer-size 1000000 apply running-config, Description: Restore the original buffer size allocated to the QoS queues to prevent any potential impact on the network.